

Estimating the Effort to Develop Screen Mockups

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Abstract—There is empirical evidence supporting the usefulness of screen mockups in improving the comprehension of use cases. Though a cost-benefit analysis would require an estimate of the effort required to produce them. In this paper, we present an empirical study with students and professionals to estimate the effort to develop mockups starting from use cases. We adopted StepWise Linear Regression to build estimation models based on use case size measures (e.g., the number of steps and the number of characters). The leave-one-out cross validation has been adopted to assess the accuracy of our models against a baseline benchmark. The results of our empirical study show that the estimates obtained with our models are significantly better than those achieved by using the chosen baseline benchmark.

Keywords—Empirical Studies, Use Cases, Screen Mockups, Requirements Engineering

I. INTRODUCTION

To improve the comprehension of Functional User Requirements (FUR) [1], use cases are often enriched with prototypes of graphical user interfaces (screen mockups, from here on) [2]. Screen mockups are used in the requirements engineering process to reduce ambiguities of functional requirements and to improve the communication among stakeholders. Screen mockups can be drawn freehand on paper or using special conceived graphical tools (e.g., Pencil¹).

In the literature, there are only few studies on the empirically assessment of screen mockups. For example, Ricca et al. [3] presented a controlled experiment and its replication, whose goal was to investigate whether the comprehension of FURs improves when use cases are enriched with screen mockups. The results showed an improvement in the comprehension of FURs ranging from 26.4% to 46.5%. The same authors as in [3] carried out a preliminary empirical study to assess the participants' perceptions about the effort to build screen mockups [4]. The results of this study suggested that the participants to the experimentation: (1) perceive screen mockups construction as a non-onerous activity and (2) believe that this activity requires a low effort

as compared to the writing of use cases. The results of both the studies above suggest that screen mockups are a viable means in the requirements engineering field, but open an interesting point from the manager point of view: how much effort is needed to develop mockups and what an estimation model needs to predict the cost to develop screen mockups? In other words, the manager needs some tools to assess the additional cost to enrich use cases with screen mockups and to get indications on whether the adoption of screen mockups keeps into account the costs for their development.

In this paper, we present an empirical study conducted within a joint venture between four Italian universities (Università della Basilicata, Università di Genova, Politecnico di Torino, and Università di Salerno) to build estimation models for the effort² needed to develop screen mockups starting from use cases. We asked 23 participants (11 software professionals and 12 students in Computer Science) to develop, screen mockups from the use cases of four different software systems. Then, we built the estimation models applying the StepWise Linear Regression (SWLR) on information about use case size measures (e.g., the number of steps or characters). Finally, we assessed the accuracy of our models against the chosen baseline benchmark (median of the effort to build mockups) using the k-fold cross validation.

The paper is organized as follows. Related work is reported in Section II. We describe the design and the execution of our empirical study in Section III, while the presentation and the discussion of the results are in Section IV. Final remarks and future work conclude the paper.

II. RELATED WORK

There are some works addressing the problem of estimating software development effort, and different measures and techniques have been employed and compared to build prediction models (e.g., [6] [7] [8] [9] [10] [11] [12] [13] [14]). In particular, measures are used to gather information about relevant aspects of software, that can be employed to

¹<http://pencil.evolus.vn/>

²In this study, we consider the time as an approximation for the effort. For example, the cognitive and development efforts of the participants were not measured. This is customary in the effort estimation field (e.g. [5]).

manage their development. Among them, some proposals analyzed the use of the information obtained from use cases to provide early estimation of development effort. We decided to restrict the discussion of related work to those proposals since they can be considered more related to the study we reported in this paper.

Karner was the first one to propose an estimation approach that exploits use cases to predict development effort, i.e., the Use Case Points (UCP) method [15]. UCP is mainly based on the counting of the number of actors and transactions (i.e., the steps) included in use cases. The popularity of this approach is mainly due to the fact that use cases are widely employed to represent software requirements as reported in a survey conducted by Neil and Laplante [16]. The mechanical structure of UCP method moves from Function Points Analysis proposed by Albrecht and Gaffney but including two groups of adjustment factors [6]. Since then, several studies have applied UCP and extensions/adaptations of UCP have been proposed (e.g., Use Case Size Points [17], Extended Use Case Points [18], UCP modified [19], Adapted Use Case Points [20], Transactions and Paths [21]). However, some issues have been highlighted about the use of estimation methods exploiting use cases, e.g., standardization of the use case description, classification and complexity of use cases. A discussion and comparison of the different UCP based proposals mentioned above have been performed in [22] through a systematic review which employs a set of attribute-based criteria (e.g., accuracy, ease of use, factor inclusion, adaptability).

Anda *et al.* report the results of three industrial case studies performed to assess the use of UCP for effort estimation [23]. The aim of the studies was to provide suggestions and guidelines to software company that intend to apply use cases for effort estimation. The results of these case studies confirmed that use cases can be profitably exploited to estimate software development effort and highlighted that the use case representation can have a strong impact on the achieved estimates.

Ochodek *et al.* investigated the application of UCP method by focusing on two sets of adjustment factors related to the environmental and technical complexity of a project [24] with the aim of simplifying UCP. The analysis considered two variants of UCP (with and without unadjusted actor weights) and the possibility of reducing the number of adjustment factors. The result of the study revealed that there was no difference in the accuracy achieved with the two variants and the number of adjustment factors could be reduced. Moreover, the conclusion was that UCP method could be simplified by just counting the total number of steps in use cases.

As for the estimation techniques employed, UCP have been mainly applied in combination with Multiple Linear Regression Model (e.g., [25]), Analogy Based Estimation (e.g., [26]), and Artificial Neural Network [25].

III. EXPERIMENTATION

We present the design underlying our empirical investigation following the guidelines provided by Wohlin *et al.* in [27]. For replication purposes, the raw data of our empirical study are available on the web³.

A. Definition and Context

The goal of our study, using the Goal Question Metric (GQM) template [28], can be summarized as follows:

Analyze: the development of screen mockups starting from use cases

For the purpose of: building a prediction model

With respect to: the effort

From the point of view of: researchers and requirements analysts

In the context of: small/medium sized software projects and young developers and professional software engineers.

The accuracy of the estimations has been compared against the one achieved using a baseline benchmark. In particular, when a requirements analyst has to predict the effort required to develop a new screen mockup he/she can simply consider the median (MedEFH) of the time expressed in minutes to develop screen mockups in the past. We chose this baseline because it is widely used in the effort estimation field (e.g., [5]), it is relatively easy to compute, and there are, as far as we know, no other mockup development effort estimation models available in the literature.

Based on the above goal and decision we formulate the following research question:

RQ. *Are the estimates based on the prediction models built using use case size measures better than those provided by the baseline benchmark when predicting the effort to develop screen mockups?*

To investigate this research question, we asked 23 participants to develop screen mockups from use cases. One or more screen mockups could be developed from the same use case. The use cases were different each other. The sample is constituted by 12 students (4 Bachelor and 7 Master) in Computer Science at the University of Basilicata and 11 young practitioners (7 software developers, 2 system managers, 1 software consultant, and 1 software tester). All the students passed a basic Software Engineering course before participating to our study. All the students can be considered the next generation of young professional developers [29]. The professional developers have been working for software companies of the contact network of one of the authors. This network was mostly created from previous research projects and also included companies that: (i) host students from the University of Basilicata for external stages or (ii) employ people, who took a Master or a Bachelor degrees at that university. The years of working experience ranged from 1

³<http://www.unisa.it/docenti/gravino/ricerca/report/effortmockup>

Use Case: Registering a Supply
Level: User-Goal
Intention in Context: The Secretary wants to record the supply of products. The products are always paid in cash upon their arrival.
Primary Actor: Secretary
Main Success Scenario:
 1. The Secretary provides as the input the number of boxes and the type of products to be bought.
 2. The system shows how much money are available in the cash (i.e., the daily availability of the cash).
 3. If there is enough money, the system confirms the supply and shows the current situation of the available products.
 4. The use case ends successfully.
Extension:
 3.a. If there is not enough money, the system informs the Secretary that the required operation cannot be accomplished. The use case fails.

Figure 1. A use case of the LaTazza system

to 3. As for the educational qualification, 3 participants had a high school degree, 5 had a Bachelor degree, and 3 had a Master degree.

B. Planning

Our study uses past information to build models for predicting the effort for developing screen mockups. Given size measures of the related use cases from the past we correlate them with the effort needed to develop screen mockup. We postulated that the effort to develop the connected screen mockup(s) should be proportional, on a first approximation, to the number of: characters and words in the use case, steps in the main success scenario, and extension points in the flow of events. This is the reason why we considered these measures to predict the effort to build screen mockups from use cases.

Figure 1 shows, as an example, one of the use case employed in our study. The use cases is specified according to the SWEED template⁴. The number of characters (spaces included) is 736, while 118 are the words. The steps in the main success scenario are four. Only one extension point (3.a) is present in that flow. Figure 2 shows a screen mockup concerning the use case in Figure 1. The shown mockup is in Italian because it has been developed by one of the participants to the study presented here.

C. Experimental Objects

We asked the participants to develop screen mockups from the uses cases of the following four desktop applications:
EasyCoin. A software application for cataloging collections

⁴<http://www.uml.org.cn/requirementproject/pdf/re-a2-theory.pdf>

Tipo	Quantità Scatole	Costo
Caffè	2	155 €
Caffè Arabica	1	77,5 €
The	1	77,5 €

Saldo Cassa: 200 € **Acquista**

Tipo	Quantità Bustine
Caffè	390
Caffè Arabica	150
The	60
Camomilla	37

Figure 2. Example of LaTazza screen mockup

of coins;

AMICO. An application intended to help Property Managers in the management of condominiums;

EasyMark. An application to manage course information about students and exam scheduling in a university;

LaTazza. An application to support the management of a coffee maker.

We used these systems because of the availability of their requirements specification documents, which included use cases compliant with the SWEED template. In addition, some of these systems have already been successfully used in empirical studies similar to the one presented here (e.g., [3], [4]). The specification documents of these systems were realized by one of the authors and used in four editions of a Software Engineering course at the University of Genova.

D. Selected Variables

The independent variables are the four use case size measures mentioned in Section III-B and summarized in Table I. We chose these measures because they are simple to collect and easy to use. The dependent variable is EFH and denotes the time (in minutes) a participant spend to develop one or more screen mockups related to a use case.

E. Execution

Before the actual experimentation, each participant underwent an individual training session. The training required to

Table I
VARIABLES EMPLOYED IN THE ANALYSIS

Measure	Description
#chars	the number of characters in the use case
#words	the number of words in the use case
#steps	the number of steps in the flow of events
#exts	the number of extensions in the flow of events
EFH	the effort required to define and design the screen mockups by analyzing the use case

develop two screen mockups using an open source computer-drawn tool (i.e., Pencil). All the participants followed the same experimental procedure: (i) downloading and install the Pencil tool and (ii) reproducing two sample mockups.

With respect to the actual experimentation, the participants accomplished the following three tasks:

- 1) *Filling in a pre-questionnaire.* It was given to the students and the professionals. The goal of each questionnaire was to collect demographic information about the participants and previous working experience.
- 2) *Developing screen mockups.* The participants were provided with a use case and were asked to develop one or more mockups. We randomly selected use cases with a number of steps from 3 to 8 and a number of extensions from 1 to 5 from the requirements specification documents of the systems described in Section III-C. The selected use cases are different each other because prediction models are built considering how the value of the dependent variable (e.g., EFH) varies with respect to the independent ones (e.g., #chars and #words). A supervisor took the time the participants spent to accomplish this task.
- 3) *Filling in a post-questionnaire.* The participants were asked to fill in a post-questionnaire. It was intended to get qualitative insights about the task *developing screen mockups* and the tool used to develop the mockups (see Table II).

To meet the student academic needs, a single experimental session was organized for each student involved. Similarly, we organized a session for each professional. To avoid biasing the results, the experimental sessions were carried out under the control of the same supervisor. Finally, the participants were not informed on the research question of our empirical study.

F. Data Analysis

For each participant, we used the information on the time spent to perform the task *developing screen mockups* and the size measures of the use cases. Then, we built the estimation models with the StepWise Linear Regression (SWLR) method and carried out the validation of the resulting LR model using the evaluation criteria described later. To perform the data analysis, we used the R statistical package⁵.

⁵<http://www.r-project.org/>

Table II
THE POST-EXPERIMENT SURVEY QUESTIONNAIRE

ID	Questions
Q1	I had enough time to complete the task:
Q2	The used experimental documentation was clear:
Q3	I found useful the execution of the task:
Q4	The ease of use of the Pencil tool is:
Q5	Finding the widgets (e.g., combo box) in the Pencil's palette is:
Q6	The use of standard drawing operations (e.g., aligning, ordering, scaling and rotating) in the Pencil tool is:
Q7	Specifying the features of the widgets (e.g., color) is:
Q8	The complexity of the screen mockups you developed is:
Q9	The effort to develop screen mockups with respect to the effort to develop use cases is:
Q10	Adding existing use cases with screen mockups is:
Q11	I believe that adding use cases with screen mockups improves the comprehension of FURs:
Q12	How do you judge your experience in the development of screen mockups?
Possible Answers	
Q1, Q2, Q3, Q11 → 1=strongly agree, 2=agree, 3=not certain, 4=agree, 5=strongly agree	
Q4, Q8, Q12 → 1=very low, 2=low, 3=medium, 4=high, 5=very high	
Q5, Q6, Q7 → 1=very difficult, 2=difficult, 3=medium, 4=easy, 5=very easy	
Q9 → 1=very low, 2=low, 3=uncertain, 4=high, 5=very high	
Q10 → 1=very burdensome, 2=burdensome, 3=uncertain, 4=not much burdensome, 5=not burdensome	

1) *Estimation technique:* Linear regression explores the relationship between a dependent variable (EFH in our study) and one or more independent variables (i.e., #chars, #words, etc...). This technique fits a model described by an equation:

$$y = b_1x_1 + b_2x_2 + \dots + b_nx_n + c$$

The dependent variable is y , while $x_1, x_2, \dots, x_n$ are the independent variables, b_i is the coefficient that represents the amount variable y changes when variables x_i changes 1 unit, and c is the intercept.

With respect to the model construction, SWLR allows computing linear regression in stages in which the choice of the independent variables from a pool of potential explanatory variables is performed by an automatic procedure. These variables can be chosen applying three approaches: forward, backward, or a combination of both. The forward approach starts with no independent variables in the model, trying out the variables one by one and including them in the model if they are statistically significant correlated with the dependent variable. The backward approach starts with all the independent variables in the model and tests them one by one. The independent variables that are not statistically significant correlated with the dependent variable are removed from the model. In the study presented here, we used a combination of the forward and backward approaches.

To evaluate the goodness of fit of a regression model, several indicators can be considered. Among them, we can exploit the square of the linear correlation coefficient (i.e., R^2). It shows the amount of variance of the dependent variable explained by the model related to the independent

variables. A good model should have an R^2 value as much as possible close to 1.

2) *Validation method*: We performed a k -fold cross validation to assess and compare the effectiveness of effort predictions achieved with the LR model. The k -fold cross validation is widely applied to assess how the results of a statistical analysis can be generalized to an independent data set [30]. In particular, when the goal is the prediction, the k -fold cross validation is used to estimate how accurately a predictive model will perform in practice. The validation goes through k rounds. Each round of the validation involves the splitting of the original data set into the training and test sets. The training set is used to build the prediction model, while the test set is exploited to validate this model. The results are averaged over the rounds. We used a leave-one-out cross validation using $k = n$, where n is the size of the original data set. Therefore, the original data set is divided into n different subsets of training and test sets, with each test set containing one observation [31].

As far as the baseline benchmark (i.e., MedEFH) is concerned, the estimated effort for the observations in the test set is obtained by considering the median of the efforts of the observations in the training set.

3) *Evaluation criteria*: To evaluate the accuracy of the predictions, we employed descriptive statistics of the absolute residuals (i.e., $|Actual\ Effort - Predicted\ Effort|$) such as Min, Max, Mean, Median (Med), and Standard Deviation (St.Dev). We also analyzed the correlation between the actual and predicted efforts by using the Kendall's tau [32] and Spearman's rho statistical tests [33]. For each observation, the less the value of the absolute residual, the better the accuracy of the obtained prediction is. The results of Kendall's tau and Spearman's rho tests range from -1 to $+1$, where $+1$ indicates perfect correlation and -1 a perfect inverse correlation. No correlation is indicated by 0. The Kendall's tau and Spearman's rho tests also return a p-value representing the significance of the correlation between the actual and predicted efforts.

To compare the accuracy of the predictions obtained with our models with the one achieved by the baseline benchmark, we also applied the Wilcoxon test. This test allows verifying whether the absolute residuals obtained with SWLR models are significantly less (i.e., better) than those achieved with MedEFH.

For all the statistical tests performed, we decided to accept a probability of 5% of committing a Type-I-Error [27].

IV. RESULTS AND DISCUSSION

Some descriptive statistics of the use case size measures and of the dependent variable EFH are shown in Table III. It is interesting to note that professionals and students, on average, spent the same time to develop screen mockups (see the row EFH in Table III).

Table III
DESCRIPTIVE STATISTICS OF THE VARIABLES

Data set	Variable	Min	Max	Mean	Med	St.Dev.
Professionals	#chars	654	1093	800.8	805.0	135.8
	#steps	3	7	5.3	5.0	1.4
	#exts	1	4	1.9	2.0	1.0
	#words	77	177	119.2	114.0	31.1
	EFH	5	48	21.1	15.0	15.2
Students	#chars	585	1162	790.2	739.0	195.9
	#steps	3	8	5.0	5.0	1.8
	#exts	1	5	2.0	1.5	1.3
	#words	74	187	114.1	114.0	31.1
	EFH	6	43	21.2	17.5	12.9

Table IV
VALIDATION METHOD RESULTS

Data set	Variable Employed	R^2	Spearman's rho		Kendall's tau	
			statistic	p-value	statistic	p-value
Professionals	#chars #exts	0.853	0.900	< 0.01	0.745	< 0.01
Students	#chars #exts	0.871	0.895	< 0.01	0.727	< 0.01

A. Results

For each data set (i.e., professionals and students), Table IV shows the variables of the built estimation models and some results about the validation of these models (i.e., the R^2 values and the results of the Spearman's rho and Kendall's tau tests). To apply SWLR, we first verified the following assumptions: the existence of a linear relationship between the independent and the dependent variables (linearity), the constant variance of the error terms for all the values of the independent variable (homoscedasticity), and the normal distribution of the error terms (normality). Since the homoscedasticity and normality assumptions of residuals were not verified, we used the natural log transformation to approximate the values to a normal distribution [34].

The obtained LR models are characterized by high R^2 values (> 0.8). For both data sets, the measures selected as best effort predictors are #chars and #exts (i.e., two out of the four variables in our original predictors' pool). Moreover, the Spearman's rho and Kendall's tau tests revealed that the predicted efforts are significantly correlated with the actual efforts. Thus, we can conclude that both the models provided a good estimate for the effort to develop screen mockups from use cases.

Descriptive statistics of the absolute residuals obtained with the leave-one-out cross validation are shown in Table V. The results show that our models outperform the baseline benchmark (we remind that: the less the value of the absolute residual, the better the accuracy of the obtained prediction is). For the students and professionals, the results are very close. However, the standard deviation value is lower on the student data set, so showing that the residuals for the associated model tend to be closer to their mean.

The results of the Wilcoxon test (SWLR vs MedEFH) are both significant for professionals (p-value=0.034) and stu-

Table V
DESCRIPTIVE STATISTICS OF ABSOLUTE RESIDUALS

Data set	Estimation Technique	Min	Max	Mean	Med	St.Dev.
Professionals	SWLR	0.25	26.33	7.88	4.36	8.30
	MedEFH	1.00	33.50	12.36	10.50	11.05
Students	SWLR	0.63	10.44	5.94	5.44	3.08
	MedEFH	1.00	26.00	11.20	11.00	8.97

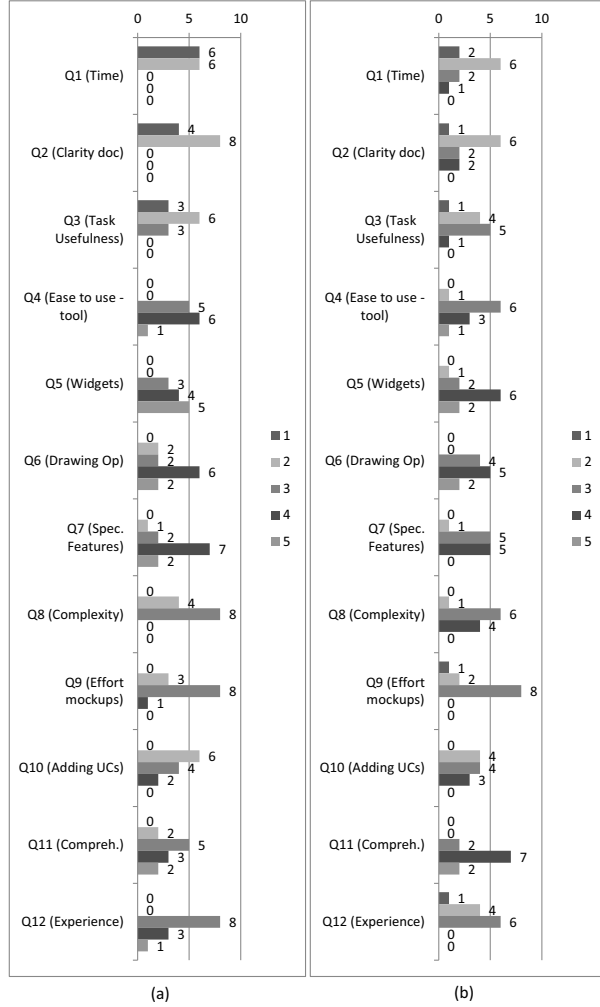


Figure 3. Post-experiment survey questionnaire (a - students) and (b - professionals)

dents (p -value=0.046). This means that the absolute residuals obtained by applying our models are significantly smaller than those achieved with MedEFH, i.e., the accuracy of predictions based on the model built with SWLR are better.

B. Post-experiment survey questionnaire

The answers to the post-experiment survey questionnaire are graphically summarized in Figure 3.a (students) and Figure 3.b (professionals) by means of bar-charts (see Table II to link the bar-charts with their respective questions).

The analyses of these answers suggest that the greater part of the students and professionals considered appropriate the time to conduct the experiment (Q1). They clearly understood the documents (Q2), and found useful the tasks (Q3). Students and professionals judged high or very high the ease of use of the Pencil tool (Q4). The answers to Q5, Q6, and Q7 revealed that all the participants to the study judged easy or very easy: (i) to find the widgets in the palette, (ii) the use of standard drawing operations, and (iii) the specification of widget features, respectively. The greater part of all the participants declared that the complexity of the developed screen mockups was medium (Q8). The participants did not have a clear-cut opinion on the difference between the effort to develop screen mockups and the effort to define use cases (Q9). Contrasting findings were also achieved with respect to the task of adding previously defined use cases with screen mockups (Q10): 6 students and 4 professionals perceive this task as burdensome, 4 students and 4 professionals did not have any opinion, and 2 students and 3 professionals perceived the task not much burdensome. The greater part of the participants did not agree on the fact that adding use cases with screen mockups improves the comprehension of FURs (Q11). This is in contrast with the results obtained in [4]. Finally, the students judged medium-high their experience in designing/developing screen mockups (Q12) while professionals judged it medium-low. The results of the post experiment survey questionnaire are close to those presented in [4]. The only relevant differences are that: (1) the participants in that study judged low the effort to add use cases with screen mockups, (2) indicated that this task was not much burdensome, and (3) agreed on the fact that adding use cases with screen mockups improves the comprehension of FURs. The reasons of these differences have to be better investigated in the future.

C. Discussion

The results of the data analysis suggest that the research question can be positively answered: the built models outperform the baseline benchmark. The use case size measures #charts and #exts can be considered good predictors of the effort to develop screen mockups. We believe that this preliminary result is particularly interesting from a practical point of view. We know, from a previous study [3], that the comprehension of functional requirements increases when use cases are added with screen mockups. Now, with these models, we have a way to predict with good approximation the time to add use cases with screen mockups. Finally, to close the loop, it would be worth analyzing whether the additional effort and cost to develop screen mockups is adequately paid back by a more valuable improved comprehension of functional requirements. This will be part of our future work.

Another aspect that deserves attention is the similarity between the models obtained for the two groups of partici-

pants (students and professionals). This result, if confirmed by future empirical studies, suggests that the estimations to develop screen mockups from use cases could not depend on the experience of the mockup developers. This could help experimenters to tune the models with students and then to use them with professionals.

D. Threats to Validity

The *external validity* threats are always present when exploiting data from a specific context. We mitigated this threat considering use cases from four desktop applications from different application domains. However, replications with other systems belonging to different domains are needed to confirm or contradict our results. Threats to external validity are also related to the complexity and the size of the use cases considered in the empirical study. Even if the selected desktop applications are “proof of concept” applications used in university courses, the size and complexity of the requirements specification documents are quite typical for simple desktop applications (for example, the number of use cases is 19 for AMICO and 20 for EasyCoin, and the number of screen mockups is 31 for AMICO and 32 for EasyCoin). Finally, the use of students as participants may also affect external validity [35]. However, the results obtained with students in terms of prediction models are very close to those obtained with professionals. Last but not least, the screen mockups have been produced using a specific tool (i.e., Pencil) and so the generalization of our results to other tools could not be doable.

The *conclusion validity threats* concern issues that affect the ability of drawing a correct conclusion. The used evaluation criteria allowed assessing in an objective way the effectiveness of the predictions. Furthermore, proper non parametric statistical tests (i.e., the Wilcoxon test) were used to statistically show the difference between our SWLR models and the chosen baseline model. The number of participants which is relatively small could affect the validity of the conclusions. Replications with a larger data set are needed.

Internal validity regards the used experimental procedure and the treatments or the experiences of the participants that threaten the researcher ability to draw correct inferences from the data. In this study, we did of our best to control all the possible “extraneous” factors.

A threat for the *construct validity* concerns the ability of establishing a correct operational measure for the concepts considered in the empirical analysis [36]. In our case, other independent variables influencing the time to develop screen mockups and not considered in our prediction models (e.g, the relationships between use cases such as extend, include and generalization) could exist. However, our SWLR models are characterized by high R^2 values (> 0.8).

V. CONCLUSION

We have presented an empirical study to estimate the effort needed to develop screen mockups starting from use cases. The results revealed that the predictions achieved are significantly better than those provided by the chosen baseline benchmark (median of the effort to build mockups).

Future work will be devoted to replicate our study on larger and more complex use cases of software systems from different problem domains. Furthermore, we will consider systems where one or more use cases can be related to one or more screen mockups to verify also whether this point has an impact on the results. Future work will be also devoted to build effort prediction models using machine learning methods [37]. We also plan to extend our study considering aspects related to accessibility and quality of screen mockups. As for the metrics used in our analysis, in the future we will evaluate the possibility of exploiting Cosmic Function Points as size measure of use cases [38].

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